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CODE:

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# Car Price Prediction using Linear Regression

import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Sample Car Dataset
data = {
    "Year": [
        2015, 2016, 2017, 2018, 2019,
        2020, 2021, 2022, 2023, 2024
    ],
    "Kilometers_Driven": [
        80000, 70000, 65000, 55000, 50000,
        40000, 35000, 25000, 15000, 5000
    ],
    "Engine_CC": [
        1200, 1300, 1400, 1500, 1600,
        1500, 1800, 2000, 1800, 2000
    ],
    "Price": [
        400000, 450000, 500000, 600000, 700000,
        850000, 1000000, 1200000, 1400000, 1600000
    ]
}

# Convert dictionary into DataFrame
df = pd.DataFrame(data)

print("CAR PRICE DATASET")
print("=" * 50)
print(df)

# Input Features
X = df[[
    "Year",
    "Kilometers_Driven",
    "Engine_CC"
]]

# Target Variable
y = df["Price"]

# Split dataset into training and testing data
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

# Create Linear Regression Model
model = LinearRegression()
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# Train the Model
model.fit(X_train, y_train)

# Predict Test Data
y_pred = model.predict(X_test)

# Calculate Performance
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

# Display Results
print("\nMODEL PERFORMANCE")
print("=" * 50)

print("Mean Squared Error:", round(mse, 2))
print("R-squared Score:", round(r2, 4))

print("\nActual Prices:", list(y_test))
print("Predicted Prices:", [round(price, 2) for price in y_pred])

# Predict Price for a New Car
new_car = pd.DataFrame({
    "Year": [2021],
    "Kilometers_Driven": [30000],
    "Engine_CC": [1600]
})

predicted_price = model.predict(new_car)

print("\nNEW CAR DETAILS")
print("=" * 50)
print("Year:", new_car["Year"].iloc[0])
print("Kilometers Driven:", new_car["Kilometers_Driven"].iloc[0])
print("Engine CC:", new_car["Engine_CC"].iloc[0])

print("\nPredicted Car Price: ₹",
      round(predicted_price[0], 2))

```

OUTPUT :

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CAR PRICE DATASET
=====
   Year  Kilometers_Driven  Engine_CC  Price
0  2015             80000         1200  400000
1  2016             70000         1300  450000
2  2017             65000         1400  500000
3  2018             55000         1500  600000
4  2019             50000         1600  700000
5  2020             40000         1500  850000
6  2021             35000         1800 1000000
7  2022             25000         2000 1200000
8  2023             15000         1800 1400000
9  2024              5000          2000 1600000

MODEL PERFORMANCE
=====
Mean Squared Error: 5363573407.21
R-squared Score: 0.9762

Actual Prices: [1400000, 450000]
Predicted Prices: [np.float64(1307894.74), np.float64(497368.42)]

NEW CAR DETAILS
=====
Year: 2021
Kilometers Driven: 30000
Engine CC: 1600

Predicted Car Price: ₹ 1084210.53

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